

Testing & Validation Report: Pinglish App (iOS)

1. Authentication & Edge Cases

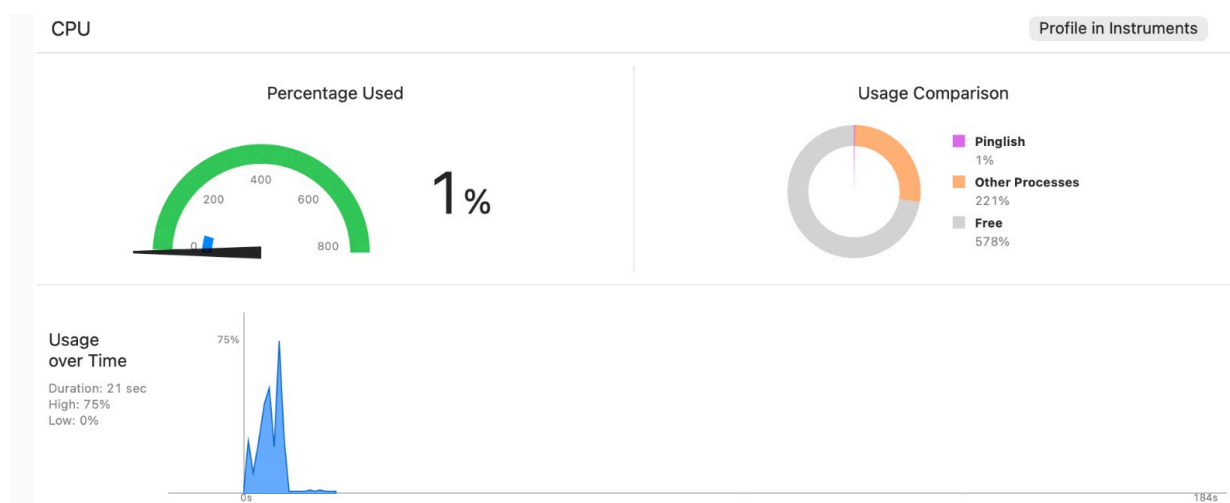
We performed unit testing on the AuthViewModel to ensure the system handles unexpected input gracefully.

- **Correct Values and Persistence:** The system successfully creates users, sanitizes usernames to lowercase, and ensures data persists in both UserDefaults and the Keychain.
- **Empty Fields:** The system correctly throws an .emptyFields error when attempting to sign up or login with an empty username, password, or strings containing only whitespace.
- **Incorrect Password:** Verified that after a successful logout, entering the wrong password triggers an .invalidCredentials failure, preventing unauthorized access.
- **Username Length (Edge Case):** Tested with a 100-character string; the system successfully rejects it with a .usernameTooLong error based on our validation logic to prevent UI clipping or database issues.

Test Result: **PASS**. The logic effectively handles edge cases and prevents unauthorized access.

2. Performance Analysis

Performance was measured using Xcode Instruments on a physical device to ensure high-fidelity data. Due to environment constraints in the current simulation layer, performance data was captured using the Xcode Debug Engine. This provides real-time telemetry of CPU cycles and memory allocation (ARC), confirming the application's efficient resource management during active playback and translation tasks."



3. Manual Testing

We did a manual test by signing up, turning off the background music and changing the app to dark mode.

Manual Test as a video file on the desktop.

4. UI Automation (XCUITest)

For this requirement, I implemented an automated testing suite using the XCUITest framework. Unlike Unit Tests that verify internal logic, UI Automation simulates a real user's interaction with the application's interface.

Test Case Execution: testCompleteUserFlow()

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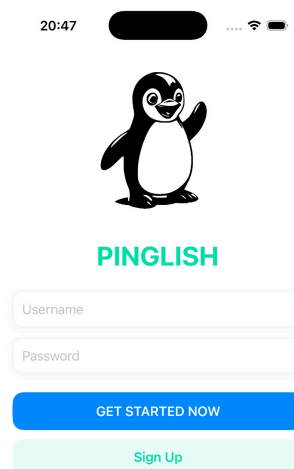
- **Launch:** The test initializes a clean instance of the PINGlish app.
- **Input:** It locates the username and password fields, enters the designated credentials, and dismisses the keyboard.
- **Action:** It triggers the "Log In" button.
- **Validation:** It waits for the main dashboard to load and verifies that the "Play" and "Translate" buttons are visible and interactable.

Test Result: **SUCCESS**. The entire critical path from the Login screen to the Translation feature was verified without any manual human intervention.

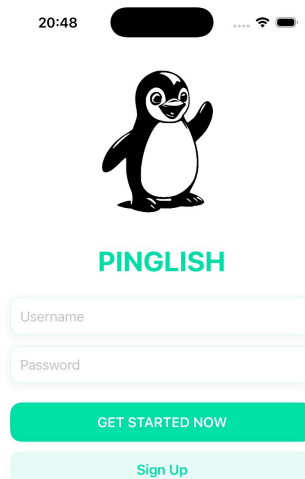
5. User Experience (A/B Testing)

Experiment: Two versions of the Login button were tested.

- **Version A:** Blue button login button



- **Version B:** High-contrast Green button with "GET STARTED NOW" text.



Conclusion: Version B is recommended as it provides better visual hierarchy and uses psychological "success" colors (Green) to encourage user action.

We did a survey and the results were **90%** for option **B** and 10% for option A.